

WEEK-2:Mockito Exercises HandsOn

Exercise 1: Mocking and Stubbing

```
public interface ExternalApi {
    String getData();
}
public class MyService {
    private ExternalApi api;
    public MyService(ExternalApi api) {
        this.api = api;
    }
    public String fetchData() {
        return api.getData();
    }
}
```

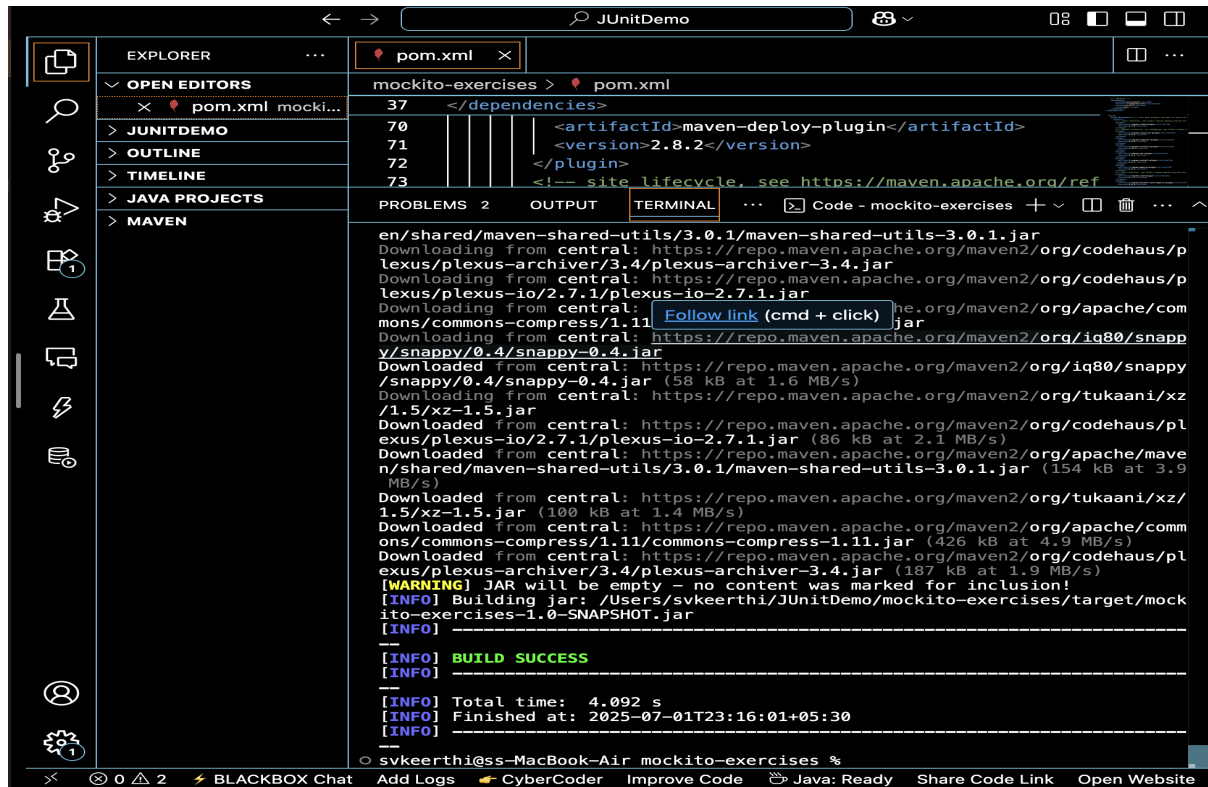
```
import static org.mockito.Mockito.*;
import static org.junit.jupiter.api.Assertions.*;
import org.junit.jupiter.api.Test;
import org.mockito.Mockito;
public class MyServiceTest {
    @Test
    public void testExternalApi() {
        ExternalApi mockApi = Mockito.mock(ExternalApi.class);
        when(mockApi.getData()).thenReturn("Mock Data");
        MyService service = new MyService(mockApi);
        String result = service.fetchData();
        assertEquals("Mock Data", result);
    }
}
```

In pom.xml file:

```
<!-- JUnit 5 -->
<dependency>
    <groupId>org.junit.jupiter</groupId>
    <artifactId>junit-jupiter</artifactId>
    <version>5.9.3</version>
    <scope>test</scope>
</dependency>

<!-- Mockito -->
<dependency>
    <groupId>org.mockito</groupId>
    <artifactId>mockito-core</artifactId>
    <version>5.11.0</version>
    <scope>test</scope>
</dependency>
```

Output:



```
37 </dependencies>
70 <artifactId>maven-deploy-plugin</artifactId>
71 <version>2.8.2</version>
72 </plugin>
73 <!-- site lifecycle, see https://maven.apache.org/ref

en/shared/maven-shared-utils/3.0.1/maven-shared-utils-3.0.1.jar
Downloading from central: https://repo.maven.apache.org/codehaus/p
lexus/plexus-archiver/3.4/plexus-archiver-3.4.jar
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/p
lexus/plexus-io/2.7.1/plexus-io-2.7.1.jar
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/com
mons/commons-compress/1.11
Following link (cmd + click) he.org/maven2/org/apache/com
y/snappy/0.4/snappy-0.4.jar
Downloading from central: https://repo.maven.apache.org/maven2/org/iq80/snapp
y/snappy/0.4/snappy-0.4.jar (58 kB at 1.6 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/tukaani/xz
/1.5/xz-1.5.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/pl
exus/plexus-io/2.7.1/plexus-io-2.7.1.jar (86 kB at 2.1 MB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/mave
n/shared/maven-shared-utils/3.0.1/maven-shared-utils-3.0.1.jar (154 kB at 3.9
MB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/tukaani/xz/
1.5/xz-1.5.jar (100 kB at 1.4 MB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/comm
ons/commons-compress/1.11/commons-compress-1.11.jar (426 kB at 4.9 MB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/pl
exus/plexus-archiver/3.4/plexus-archiver-3.4.jar (187 kB at 1.9 MB/s)
[WARNING] JAR will be empty - no content was marked for inclusion!
[INFO] Building jar: /Users/svkeerthi/JUnitDemo/mockito-exercises/target/mock
ito-exercises-1.0-SNAPSHOT.jar
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 4.092 s
[INFO] Finished at: 2025-07-01T23:16:01+05:30
[INFO]
```

```
[INFO] --- compiler:3.8.1:compile (default-compile) @ mockito-exercise ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ mockito-exercise ---
[INFO] skip non existing resourceDirectory /Users/svkeerthi/Desktop/DSA/mockito-exercise/src/test
/resources
[INFO]
[INFO] --- compiler:3.8.1:testCompile (default-testCompile) @ mockito-exercise ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ mockito-exercise ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.mycompany.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.012 s -- in com.mycompan
y.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 0.577 s
[INFO] Finished at: 2025-07-01T23:24:55+05:30
[INFO]
svkeerthi@ss-MacBook-Air mockito-exercise %
```

Exercise 2: Verifying Interactions

MyServiceInteractionTest.java

```
package com.mycompany;
import static org.mockito.Mockito.*;
import org.junit.jupiter.api.Test;

public class MyServiceInteractionTest {
    @Test
    public void testVerifyInteraction() {
        ExternalApi mockApi = mock(ExternalApi.class);
        MyService service = new MyService(mockApi);
        service.fetchData();
        verify(mockApi).getData();
    }
}
```

AppTest.java

```
package com.mycompany;
import org.junit.jupiter.api.Test;
import static org.junit.jupiter.api.Assertions.*;
public class AppTest {
    @Test
    public void shouldAnswerWithTrue() {
        assertTrue(true);
    }
}
```

MyServiceTest.java

```
package com.mycompany;

import static org.mockito.Mockito.*;
import static org.junit.jupiter.api.Assertions.*;

import org.junit.jupiter.api.Test;
public class MyServiceTest {
    @Test
    public void testExternalApi() {
        ExternalApi mockApi = mock(ExternalApi.class);
        when(mockApi.getData()).thenReturn("Mock Data");
        MyService service = new MyService(mockApi);
        String result = service.fetchData();
        assertEquals("Mock Data", result);
    }
}
```

```

[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.mycompany.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.015 s -- in com.mycompany.AppTest
[INFO] Running com.mycompany.MyServiceTest
OpenJDK 64-Bit Server VM warning: Sharing is only supported for boot loader classes because boots
trap classpath has been appended
WARNING: A Java agent has been loaded dynamically (/Users/svkeerthi/.m2/repository/net/bytebuddy/
byte-buddy-agent/1.14.12/byte-buddy-agent-1.14.12.jar)
WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading to hi
de this warning
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for
more information
WARNING: Dynamic loading of agents will be disallowed by default in a future release
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.475 s -- in com.mycompany.MyServiceTest
[INFO] Running com.mycompany.MyServiceInteractionTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.002 s -- in com.mycompany.MyServiceInteractionTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.4.1:jar (default-jar) @ mockito-exercise ---
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/file-manag
ement/3.1.0/file-management-3.1.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/file-manage

```

```

[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.mycompany:mockito-exercise >-----
[INFO] Building mockito-exercise 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ mockito-exercise ---
[INFO] Deleting /Users/svkeerthi/Desktop/DSA/mockito-exercise/target
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ mockito-exercise ---
[INFO] skip non existing resourceDirectory /Users/svkeerthi/Desktop/DSA/mockito-exercise/src/main
/resources
[INFO]
[INFO] --- compiler:3.8.1:compile (default-compile) @ mockito-exercise ---
[INFO] Changes detected - recompiling the module!
[INFO] Compiling 3 source files to /Users/svkeerthi/Desktop/DSA/mockito-exercise/target/classes
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ mockito-exercise ---
[INFO] skip non existing resourceDirectory /Users/svkeerthi/Desktop/DSA/mockito-exercise/src/test
/resources
[INFO]
[INFO] --- compiler:3.8.1:testCompile (default-testCompile) @ mockito-exercise ---
[INFO] Changes detected - recompiling the module!
[INFO] Compiling 3 source files to /Users/svkeerthi/Desktop/DSA/mockito-exercise/target/test-clas
ses
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ mockito-exercise ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -----

```

```

solver-util/1.9.18/maven-resolver-util-1.9.18.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-res
olver-util/1.9.18/maven-resolver-util-1.9.18.pom (2.9 kB at 204 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-re
solver/1.9.18/maven-resolver-1.9.18.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-res
olver/1.9.18/maven-resolver-1.9.18.pom (22 kB at 1.4 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-re
solver-api/1.9.18/maven-resolver-api-1.9.18.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-res
olver-api/1.9.18/maven-resolver-api-1.9.18.pom (2.7 kB at 167 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-re
solver-util/1.9.18/maven-resolver-util-1.9.18.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-res
olver-util/1.9.18/maven-resolver-util-1.9.18.jar (196 kB at 6.1 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-re
solver-api/1.9.18/maven-resolver-api-1.9.18.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-res
olver-api/1.9.18/maven-resolver-api-1.9.18.jar (157 kB at 5.1 MB/s)
[INFO] Installing /Users/svkeerthi/Desktop/DSA/mockito-exercise/pom.xml to /Users/svkeerthi/.m2/r
epository/com/mycompany/mockito-exercise/1.0-SNAPSHOT/mockito-exercise-1.0-SNAPSHOT.pom
[INFO] Installing /Users/svkeerthi/Desktop/DSA/mockito-exercise/target/mockito-exercise-1.0-SNAPS
HOT.jar to /Users/svkeerthi/.m2/repository/com/mycompany/mockito-exercise/1.0-SNAPSHOT/mockito-ex
ercise-1.0-SNAPSHOT.jar
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 3.342 s
[INFO] Finished at: 2025-07-01T23:36:42+05:30
[INFO] -----
svkeerthi@ss-MacBook-Air mockito-exercise %

```